

Infinite Dictionaries

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Codes

AI Code Tips

Description

A dictionary is a set of key-value pairs, for example:

```
{'color':'red', 'price':2, 7:'test', 100:-100}
```

As you can see, **keys and values can be strings or integers**. What's more, values can also be dictionaries or variable references. Here is the formal definition of terms that will be used soon:

key ::= INTEGER | STRING

value ::= INTEGER | STRING | dict

pair ::= key ':' value

dict ::= '{' [pair (',' pair)*] '}'

var ::= 'a'|'b'|'c'|...'z'

slot ::= var('[' key '])*

lvar ::= slot

rvar ::= slot | value

Here ('[' key '])* means zero or more subscripts, [pair (',' pair)*] means zero or more key-value pairs.

Strings are always enclosed by single quotes (") and consists of up to 10 lower-case letters. Integers always have absolute values of no more than 1000. You can insert spaces anywhere, except inside strings or integers. For example, { 'a':-1 } and { 'a' : - 1 } are the same, but { 'a b':1 } and { 'a':- 1 } are both illegal.

Your task is to execute a series of commands and print the results. There are 3 kinds of commands:

1. Assignment: <lvar> = <rval>

After assigning a slot to a slot (rather than a value), the left-hand slot will be holding a reference to the right-hand. For example, After executing the following commands, b[1][0] is 1, rather than 0:

```
a = {0:0}
```

```
b = {}
```

```
b[1] = a
```

```
a[0] = 1
```

Slots must be assigned before it is *read* or *subscripted*, and integers and strings cannot be subscripted. Consider the following command list:

```
c = {}
```

```
c[0] = 3
```

```
c[1] = c[0]
```

```
d[0] = 'i'
```

```
c = d
```

```
d = c[1]['a']
```

```
c[2][2] = 2
```

The first three commands are legal, but the next two are both illegal because slot d must be assigned before it is read or subscripted. The last three are also illegal.

2. **Length:** length(<slot>)

Output the number of key-value pairs in the slot. Note that nested pairs are not counted. For example:

```
a = {0: {0:0, 1:1}}
```

```
length(a)
```

will output 1, not 3. In this command, it is guaranteed that <slot> is storing a dictionary, not a string or an integer.

3. **Infinity test:** test(<slot>)

If the slot can be subscripted indefinitely, output 1. Otherwise, output 0. For example, after executing the following command list:

```
d = {}
```

```
d[0] = d
```

Then d is infinite, since `d[0][0][0][0][0][0]...` is always d. In this command, it is guaranteed that <slot> is storing a dictionary, not a string or an integer.

Input

The input contains at most 10000 lines of commands, each line will be non-empty and will contain no more than 300 characters. All the commands are legal.

Output

Print the output (one line for each length/test command).

Samples

input	Copy
<pre>c = {} d = {'color': 'red', 'price': 2, 7: 'test', 100: -100} length(d) d[7] = {'this': 'is', 'a': 'book'} length(d) d[8] = {'this' : 'is', 'another' : {'a' : 'book', 'b': 'book2'} } length(d) c[7] = c test(c) test(d) length(c) d[0] = c length(d) test(d[0])</pre>	
output	Copy
<pre>4 4 5 1 0 1 6 1</pre>	

Hint

The term definitions in this problem use a modified BNF grammar notation that is described in the official documentation of the Python programming language. Here is the explanation of the notation, extracted from the documentation:

Each rule begins with a name (which is the name defined by the rule) and ::= . A vertical bar (|) is used to separate alternatives; it is the least binding operator in this notation. A star () means zero or more repetitions of the preceding item; likewise, a plus (+) means one or more repetitions, and a phrase enclosed in square brackets ([]) means zero or one occurrences (in other words, the enclosed phrase is optional). The * and + operators bind as tightly as possible; parentheses are used for grouping. Literal strings are enclosed in quotes. White space is only meaningful to separate tokens. Rules are normally contained on a single line; rules with many alternatives may be formatted alternatively with each line after the first beginning with a vertical bar*

Source

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HZNUOJ is based on HUSTOJ

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